Implementation of Three Phase Inverter connected to Three Phase Balanced Grid using Hysteresis Control

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Aim & Objective

- Design and simulate a three-phase inverter connected to a three-phase balanced grid using Hysteresis control.
- Design the series inductor filter such that total harmonic distortion (THD) in current is less than 5%.
- Design the controller to have fast settling time for different P_{ref} and Q_{ref} references.

System Overview

Main components

- DC voltage source
- IGBT based three phase inverter
- Inductive filters
- Three phase balanced grid
- Hysteresis block (Relay)
- Logic gates (NOT) & Summation block
- Sensors & Display items: Voltage, Current, Display, Scopes

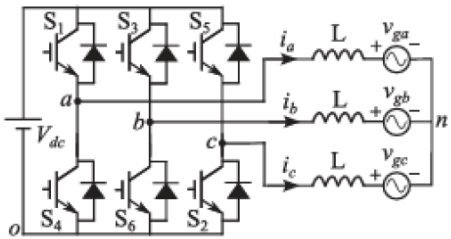


Figure: System Model

Why Hysteresis Control?

- PI control is a **linear control technique** in which we require the system frequency ω and the angle θ (angle of d-axis with respect to α-axis). These quantities are obtained using a **Phase Locked Loop (PLL)**.
- **PI** control technique require **DC** values but there is no such constraint in Hysteresis control technique.
- Hysteresis control is a **non-linear control technique** for which neither ω nor θ is required, because we do not perform the d–q axis **Park's transformation**.
- In hysteresis control, the three-phase (a, b, c) quantities are transformed into α–β axis using the **Clarke's transformation**.
- Hence, Hysteresis control technique remove the dependency on PLL and Park's transformation.

Implementation Steps

- We obtained V_α and V_β from V_a, V_b and V_c using the Clarke transformation.
- Using the governing equations, we computed $I_{\alpha\text{ref}}$ and $I_{\beta\text{ref}}$ from P_{ref}, Q_{ref}, V_α and V_β.
- Then we obtained $I_{a\text{ref}}$, $I_{b\text{ref}}$ and $I_{c\text{ref}}$ from $I_{\alpha\text{ref}}$ and $I_{\beta\text{ref}}$ using the inverse Clarke transformation.
- We calculated the error:

$$\text{Error} = I_{\text{ref}} - I$$

  and fed this error into the hysteresis block (relay).
- The hysteresis block generated the gate pulses, which were fed back to the inverter for gate control operation.

Governing Equations

Reference Current Calculation

$$I_{\alpha ref} = \frac{3}{2} \cdot \frac{V_\alpha P_{ref} - V_\beta Q_{ref}}{V_\alpha^2 + V_\beta^2}$$

$$I_{\beta ref} = \frac{3}{2} \cdot \frac{V_\beta P_{ref} + V_\alpha Q_{ref}}{V_\alpha^2 + V_\beta^2}$$

Clarke Transformation

$$\begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

Inverse Clarke Transformation

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} I_{\alpha ref} \\ I_{\beta ref} \end{bmatrix}$$

Research Paper Reference

Reference Paper:

- *"An abc Reference Frame Hysteresis Current Controller for Grid Connected Converter With Constant Switching Frequency Operation"* by Ravi Varma Chavali, Anubrata Dey, Biswarup Das.

Data Used:

Grid voltage (LL rms)	240 V
Grid frequency	50 Hz
Filter capacitor	6.25 F
Filter inductance	4.97 mH
Filter resistance	1.3
DC-link voltage	600 V

Simulation model in MATLAB SIMULINK

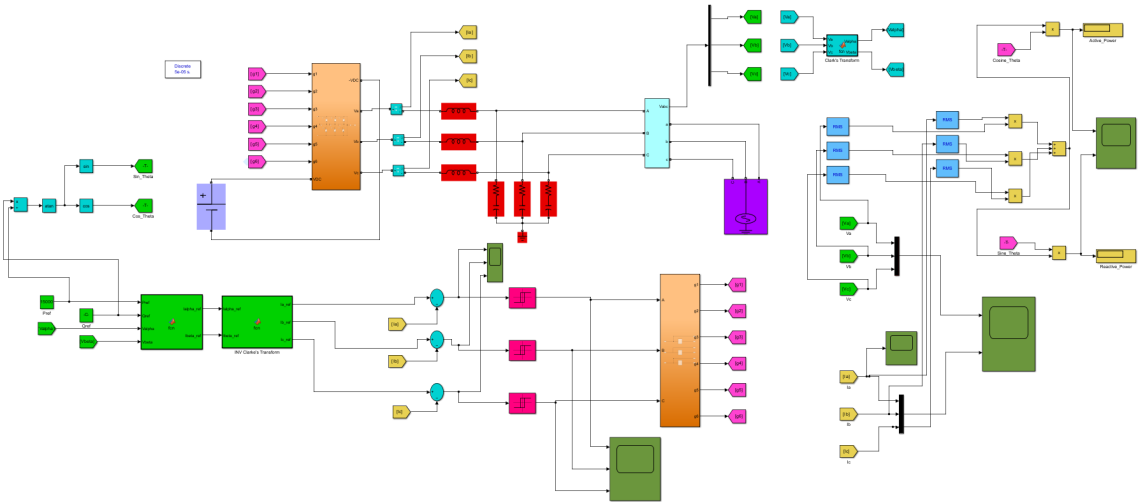
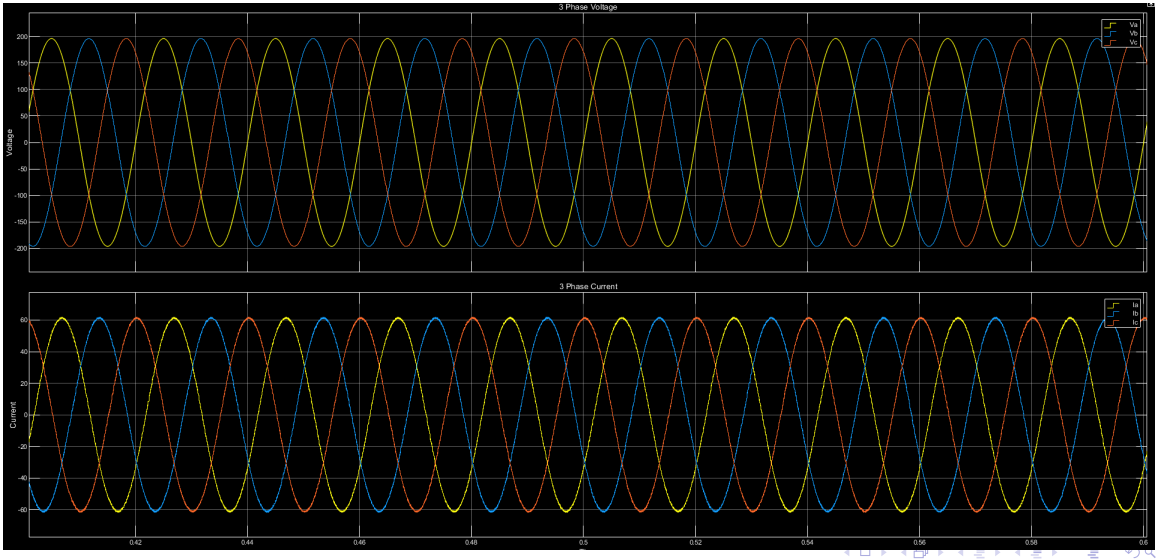
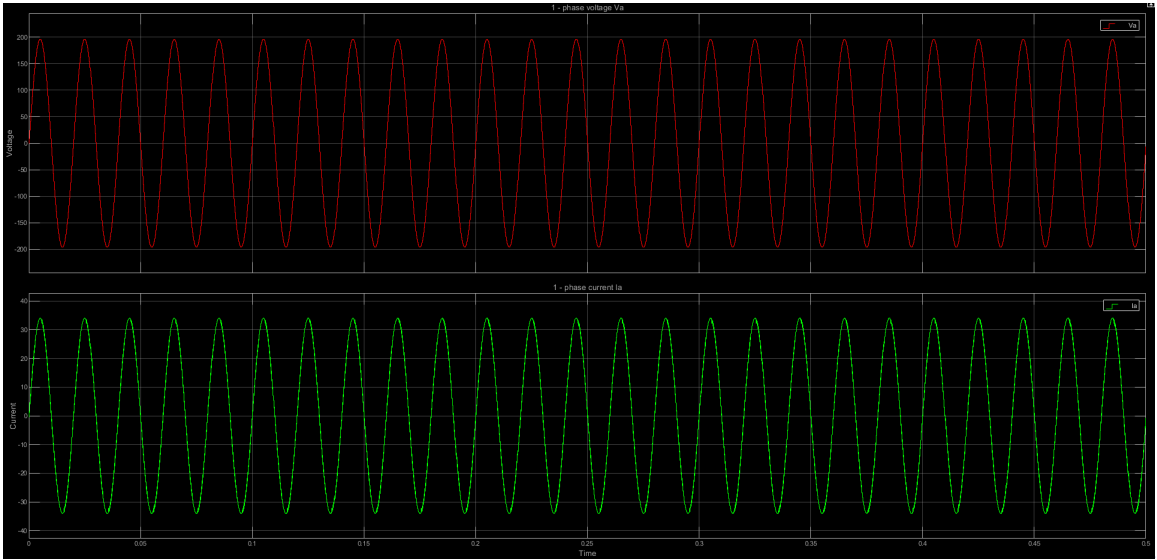


Figure: System Simulink Model

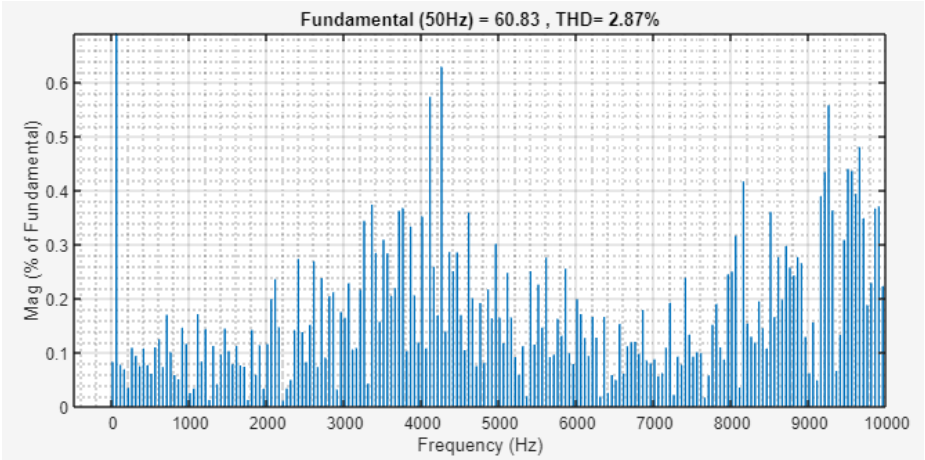
Simulation Results: 3-ϕ Voltage & 3-ϕ Current



Simulation Results: 1-ϕ Voltage & 1-ϕ Current



Simulation Results: Total harmonic distortion (THD) in current

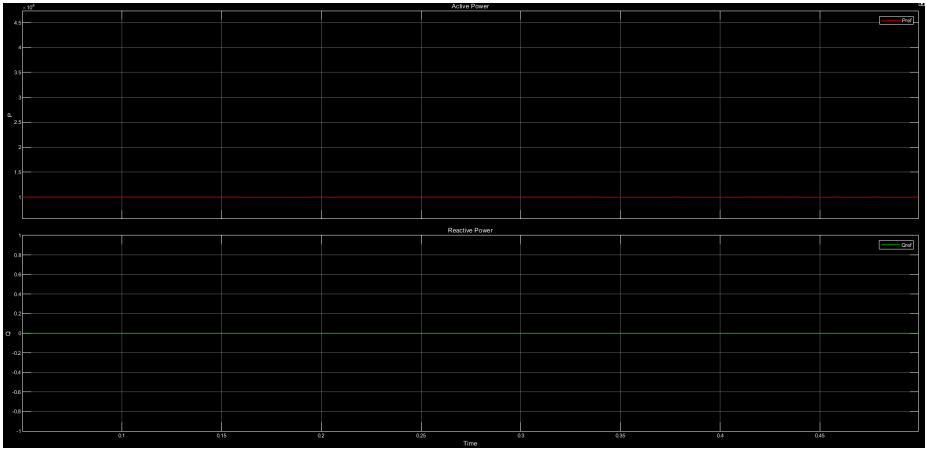


Simulation results for P_{ref} and Q_{ref} given below:

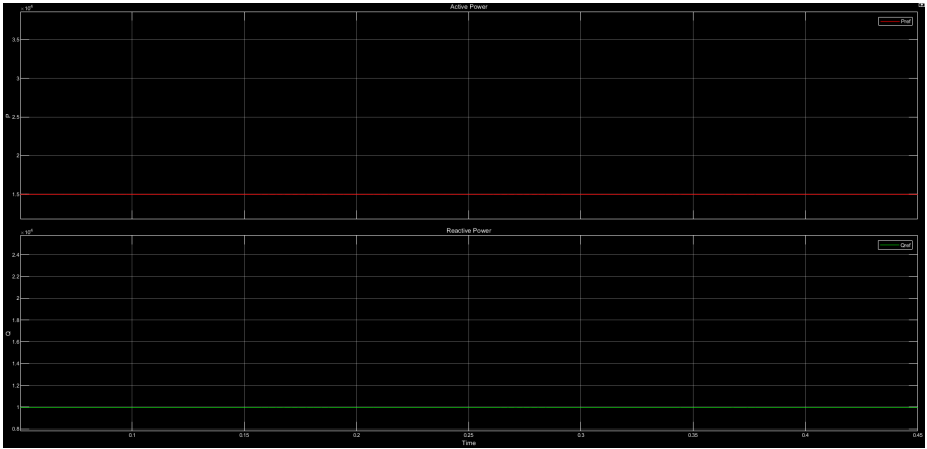
- (i) 10 kW, 0 kVAR
- (ii) 15 kW, 10 kVAR
- (iii) 8 kW, –10 kVAR
- (iv) –10 kW, 10 kVAR

Must meet references without overshoot.

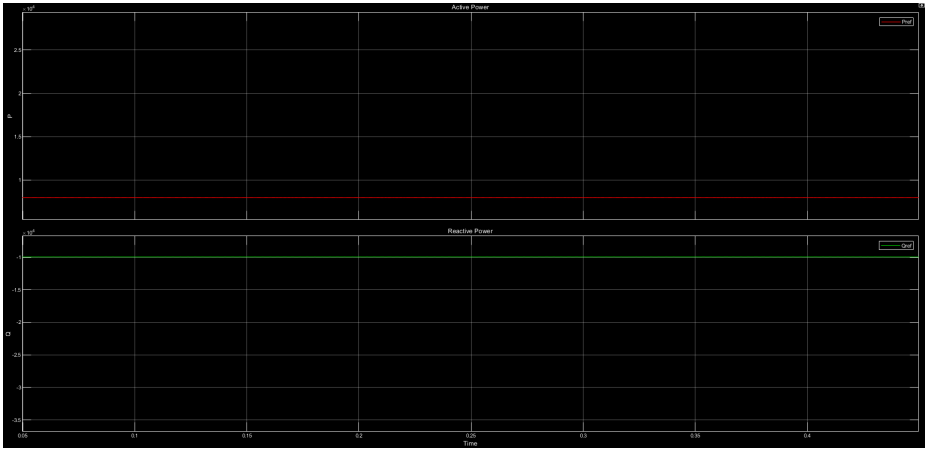
Simulation Results: $P_{ref} = 10000W, Q_{ref} = 0VAR$



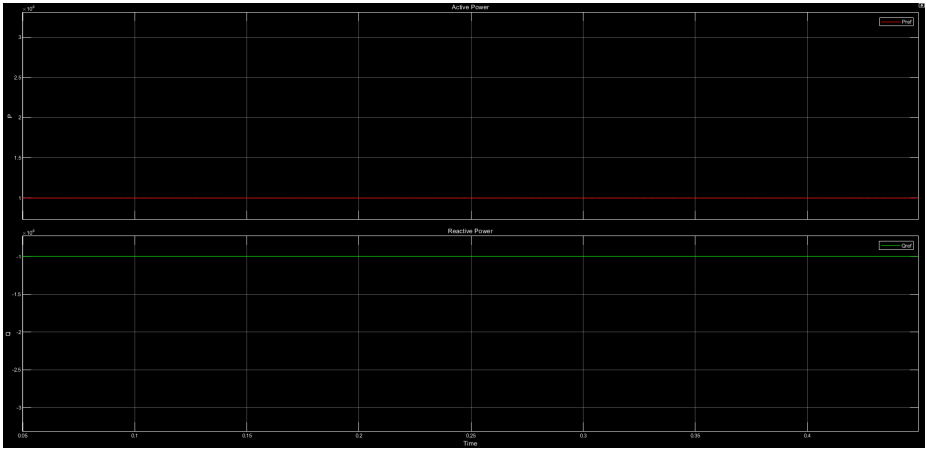
Simulation Results: $P_{ref} = 15000W, Q_{ref} = 10000VAR$



Simulation Results: $P_{ref} = 8000W, Q_{ref} = -10000VAR$



Simulation Results: $P_{ref} = -10000\,W$, $Q_{ref} = 10000\,VAR$



Conclusion

- We were able to successfully implement the Hysteresis control technique for three phase inverter connected to three phase balanced grid.
- We have successfully reduced the total harmonic distortion (THD) below 5%.
- We have successfully tracked the P_{ref} and Q_{ref} provided.
- We were able to get the Current and Voltage waveform without any overshoot and delay.

References

- "An abc Reference Frame Hysteresis Current Controller for Grid Connected Converter With Constant Switching Frequency Operation" by Ravi Varma Chavali , Anubrata Dey Member, IEEE, and Biswarup Das.
- Lecture notes: *Advanced Control Design Techniques for Power Converter* — Dr. A. V. Ravi Teja.

Thank You!